Laser Sources to Simplify Slow-and-Wide Optical Interconnects

Alan Liu

CEO, Quintessent

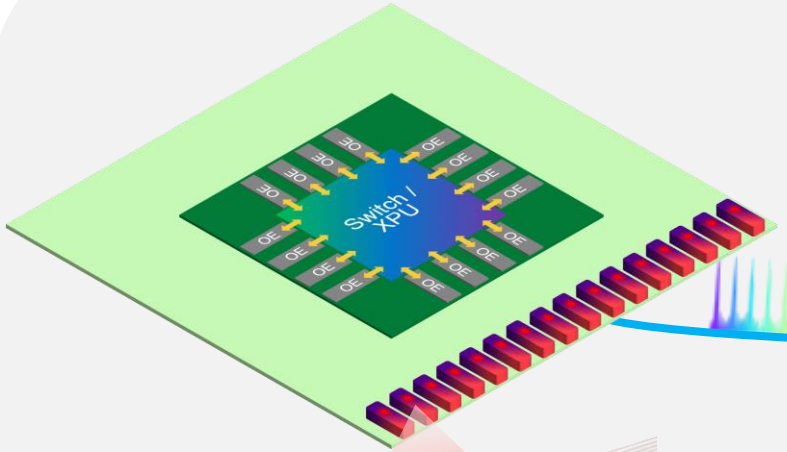


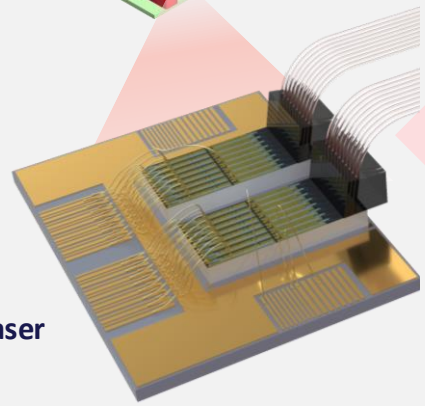
OCP FUTURE TECHNOLOGIES SYMPOSIUM

October 14, 2026
San Jose, California

One Technology Platform. Multiple Applications.

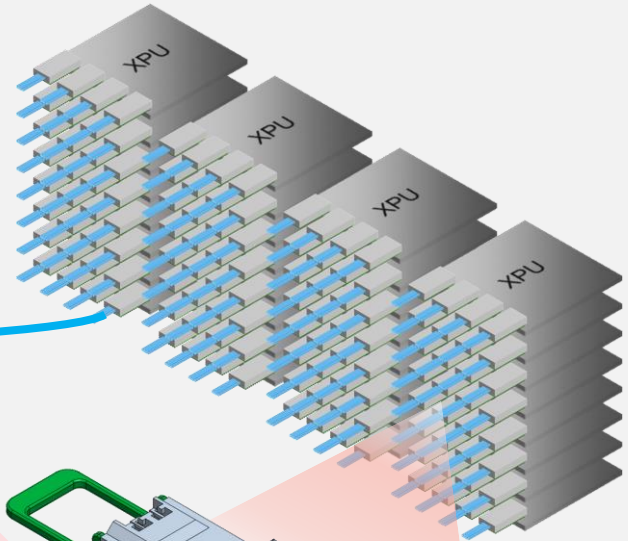
CPO / NPO

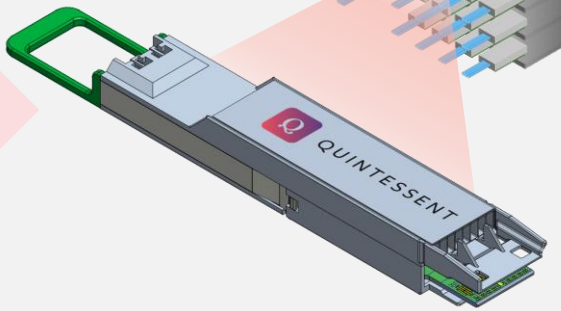




DWDM Comb Laser for CPO ELS

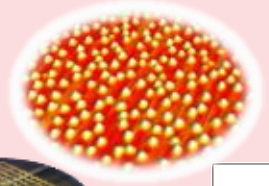
Pluggables

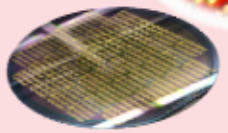




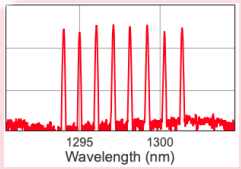
DWDM High-RAS Transceiver for Scale-Up/-Out

O-band Quantum Dot Gain





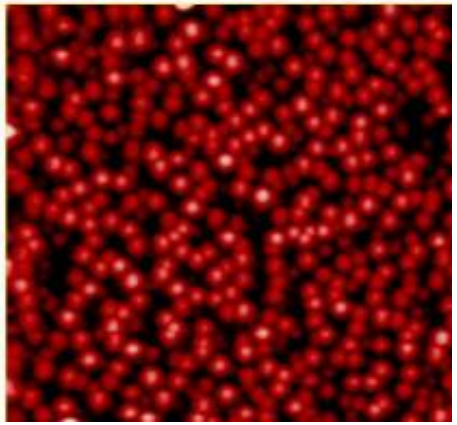
Heterogeneous integration

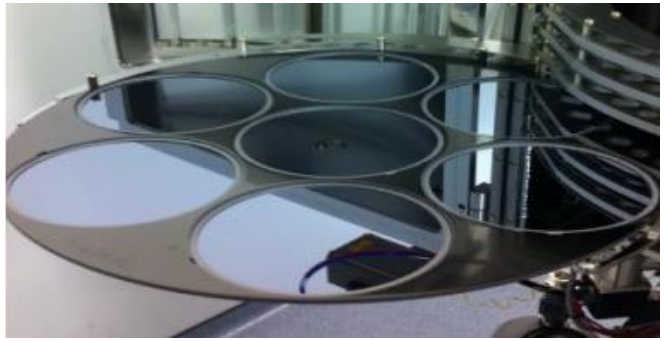


Scalable architecture

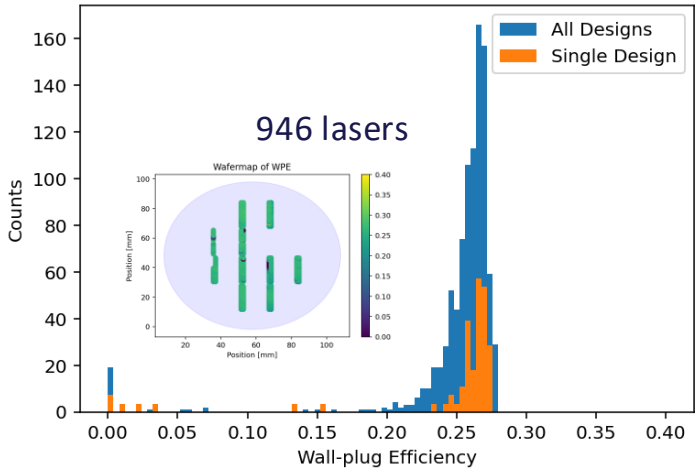
O-Band GaAs Quantum Dot Gain Foundation

- High volume o-band quantum dot epi manufacturable on up to 6” GaAs substrates
- 750+ wafers grown to date
- Results below from 100% foundry flow in EVT phase (epi, III-V processing)
- ~3000 lasers from 3 epi lots, unoptimized WPE clusters around 25-30% in every lot

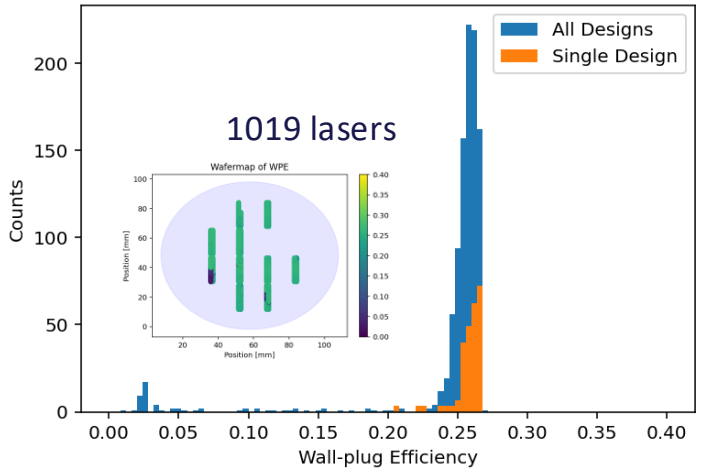




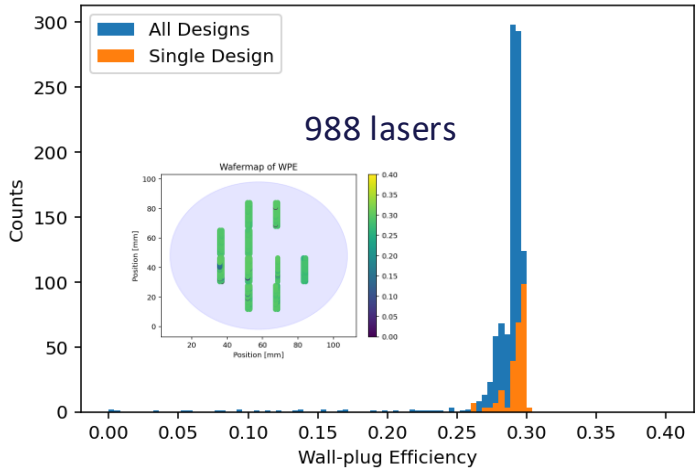
Epi1



Epi2

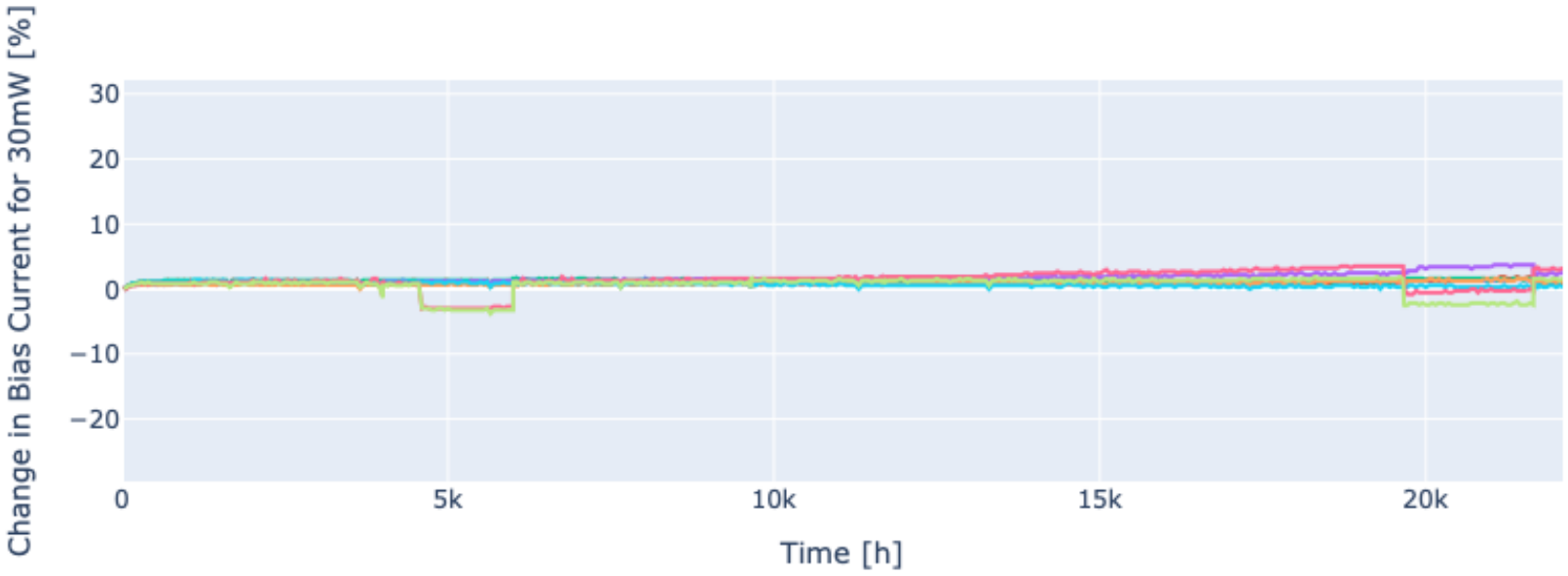


Epi3



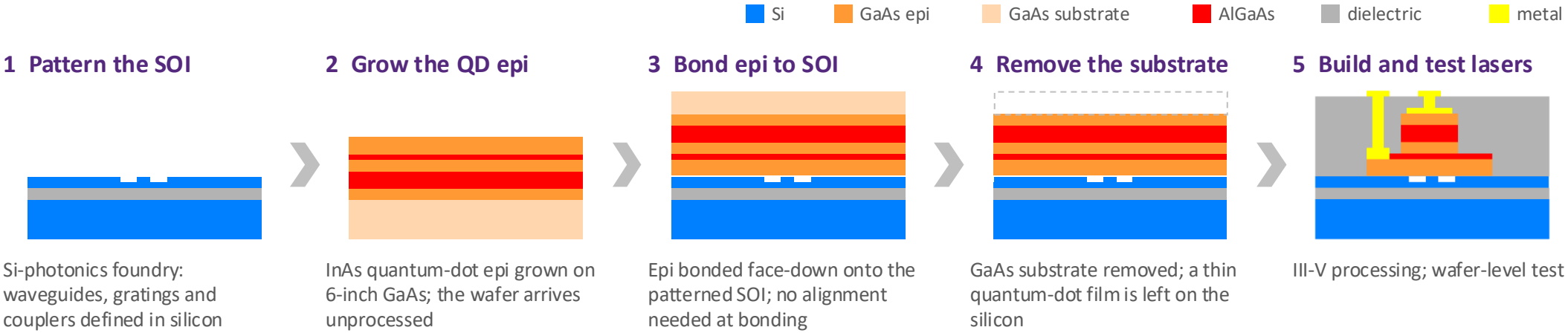
Stable Beyond >22,000h in Accelerated aging

- Preliminary reliability data – lot began aging in 2023
- Fabry-Perot quantum dot lasers aged at 80C and 125 mA
  - Realistic operating current in a system for 30 mW output power
- Now passed 22,000h and on-going
- Launched larger reliability campaign with >150 devices, with greater acceleration factors



Heterogeneous integration: O-Band GaAs quantum dots bonded onto silicon photonics

O-band GaAs quantum-dot gain wafer-bonded onto patterned SOI. The silicon defines the cavity and carries the light in silicon waveguides; the dots supply the gain — **no InP, no regrowth, no e-beam, no cleaved facets**.

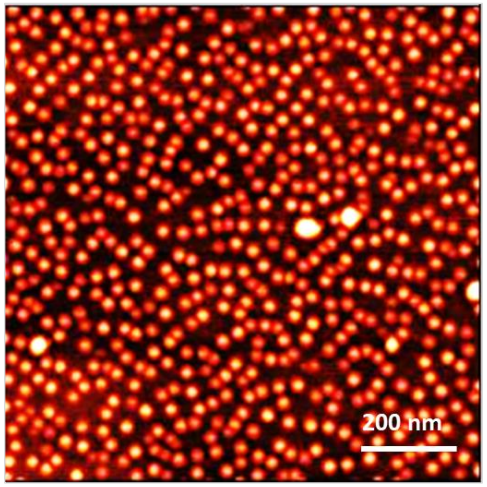


No facets → wafer level optical and electrical test of every laser on the wafer; known-good die

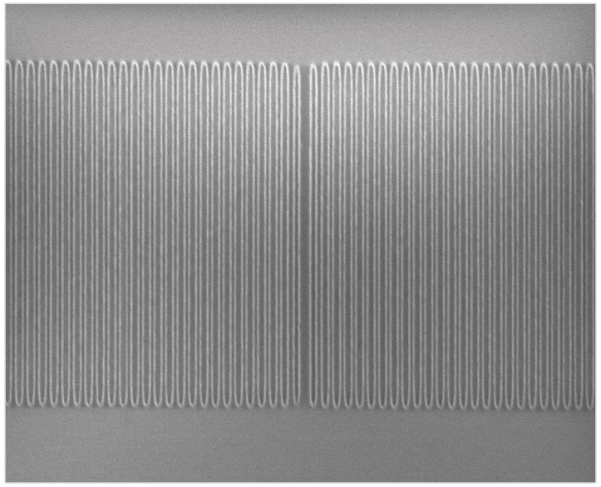
No e-beam or III-V regrowth → cavity is defined by silicon photonic lithography (65 nm or better)

No Indium Phosphide → gain material all GaAs based

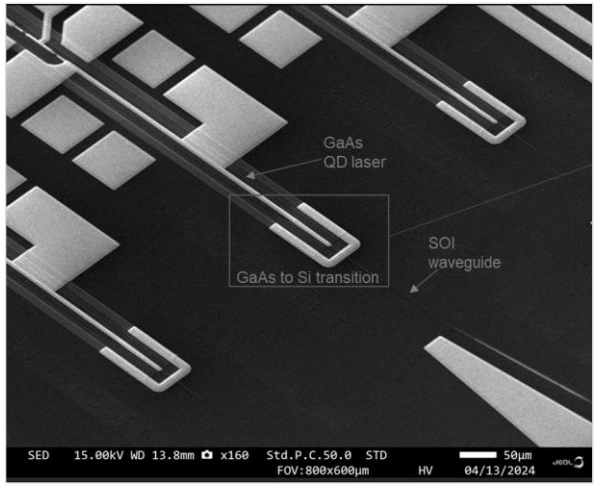
From quantum dots to packaged modules



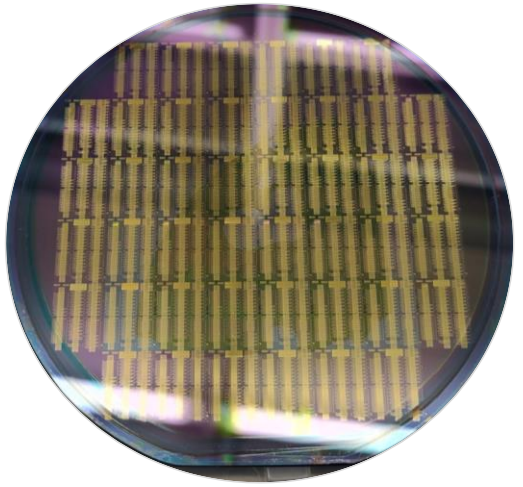
InAs quantum dots on GaAs — AFM (200 nm scale bar)



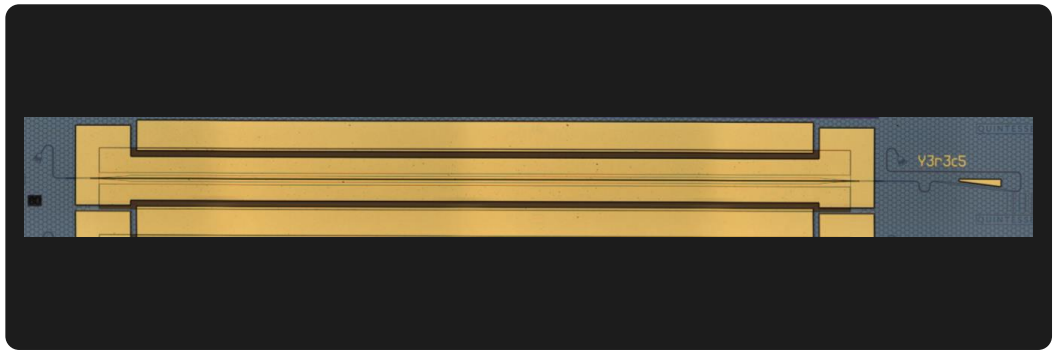
Waveguide and cavity design in silicon



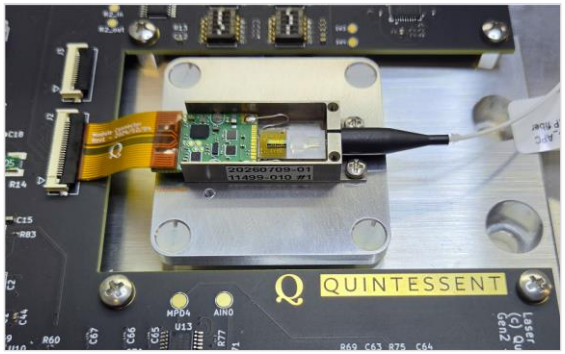
QD-on-Si lasers: GaAs QD mesas on SOI waveguides — SEM ×160 (50 µm bar)



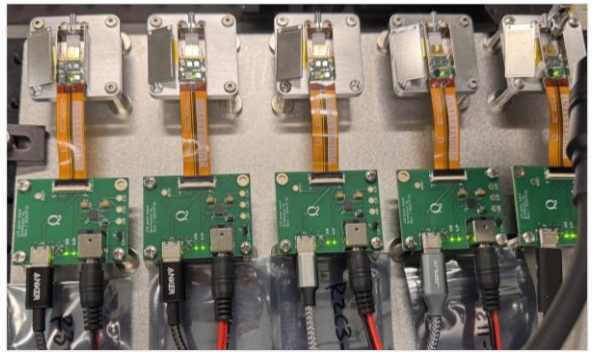
QD-on-Si laser wafer: GaAs QD lasers bonded and processed on SOI



Fully fabricated QD-on-Si lasers — top-down optical micrograph (gain sections, contacts and pads)



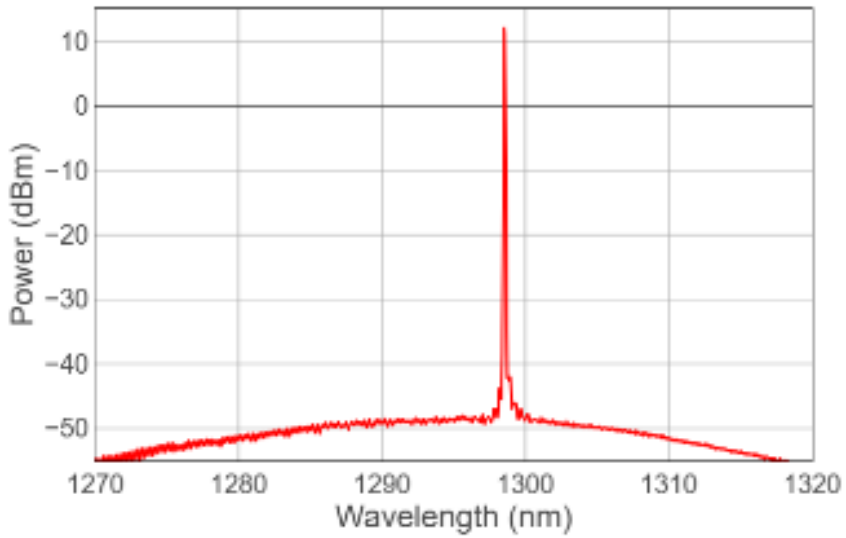
Packaged, fiber-pigtailed comb-laser module on its evaluation board



Prototype laser modules built for customer sampling

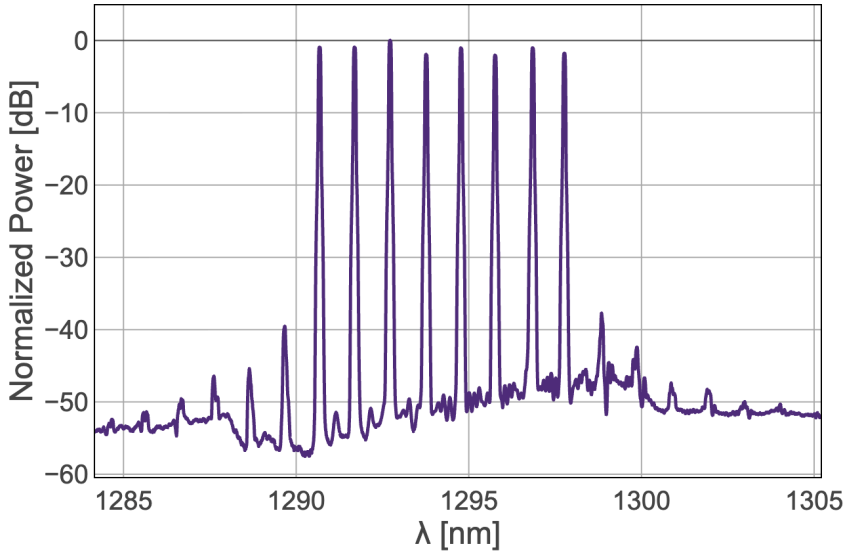
Single and multi-λ sources from the same platform

Single wavelength: QD DFB



- One laser per λ
- >55 dB SMSR, mode-hop-free vs. bias;

Multi-wavelength: QD comb laser

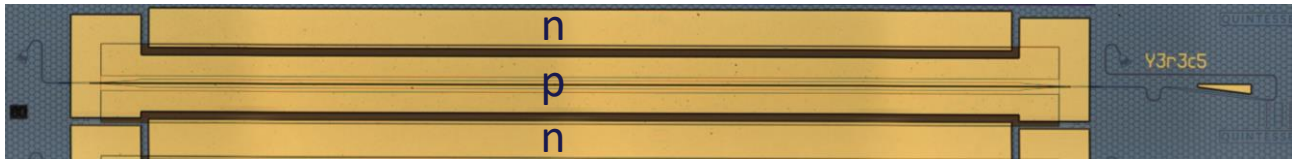


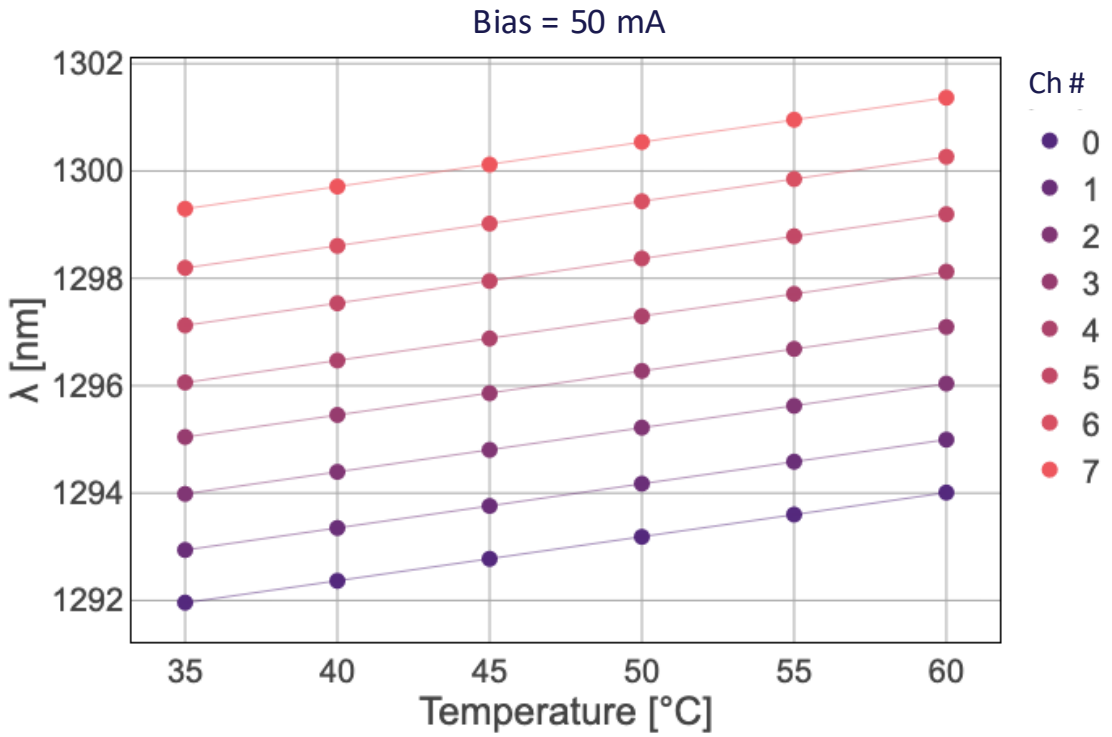
- One cavity → 8 equally spaced lines; one current source, no λ control
- Lines drift together with temperature; spacing set by the cavity

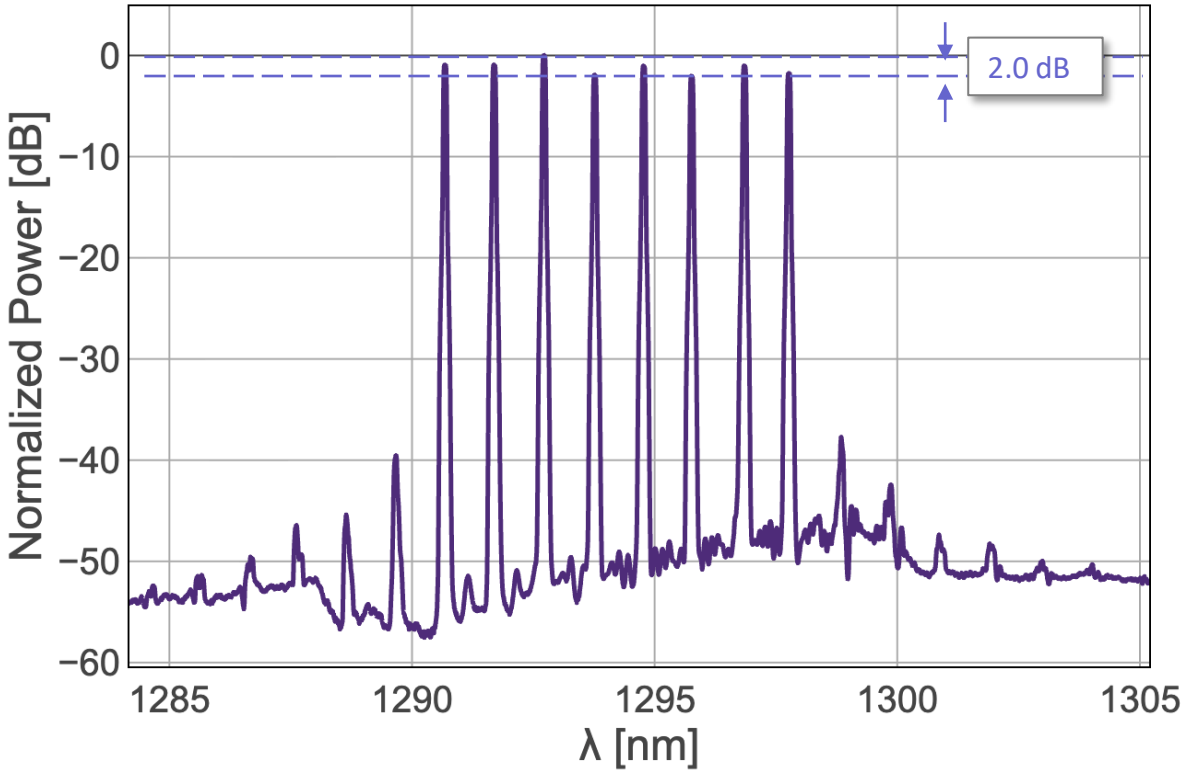
Same QD epi, same QD-on-Si process

Single Cavity, Single Bias, Multi - λ Comb Laser

Generated by a single silicon chip 3 x 0.25 mm, no feedback or control, single current source

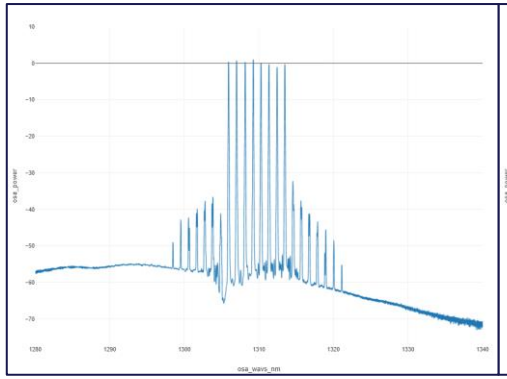




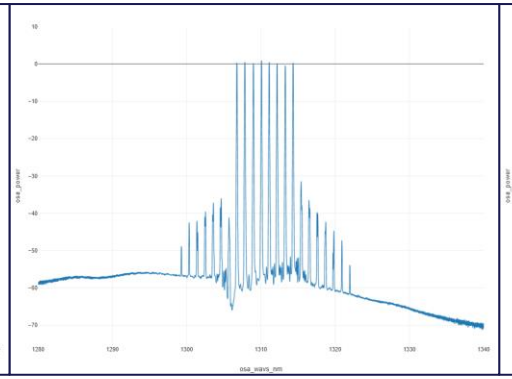


Uniform Comb from 20°C-110°C

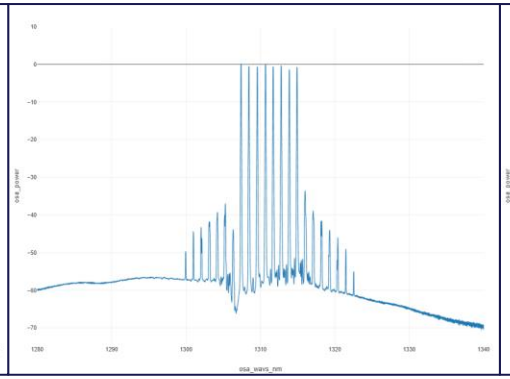
20°C



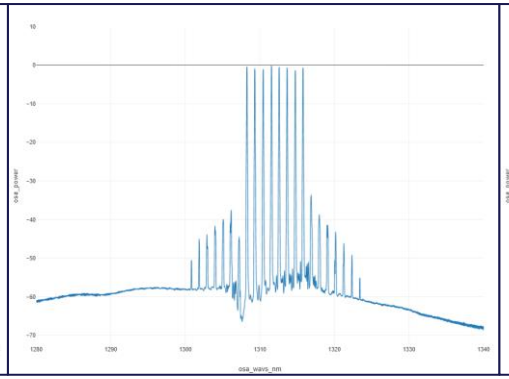
30°C



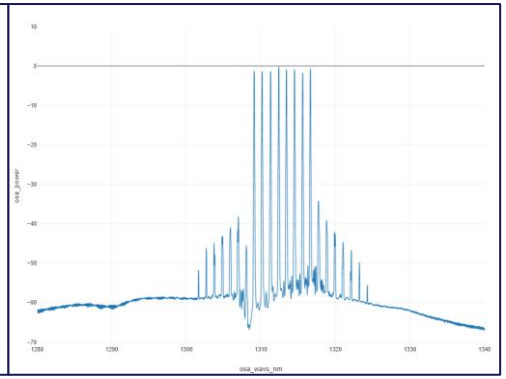
40°C



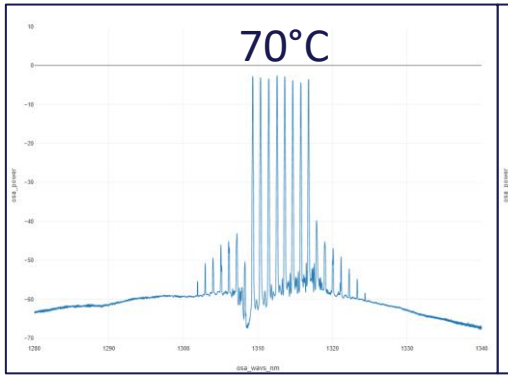
50°C



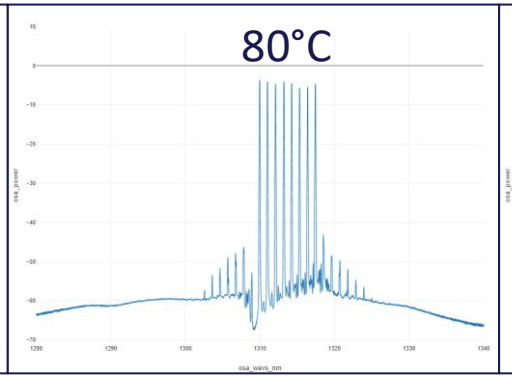
60°C



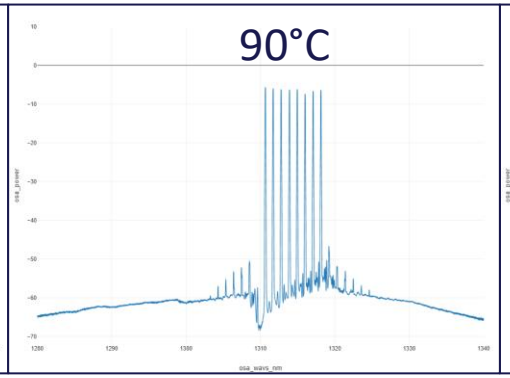
70°C



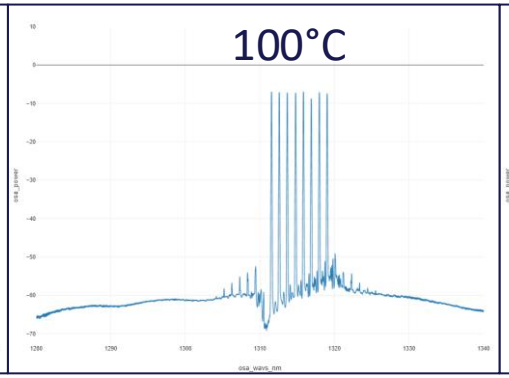
80°C



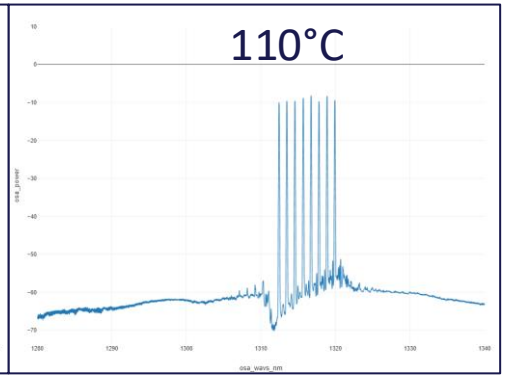
90°C



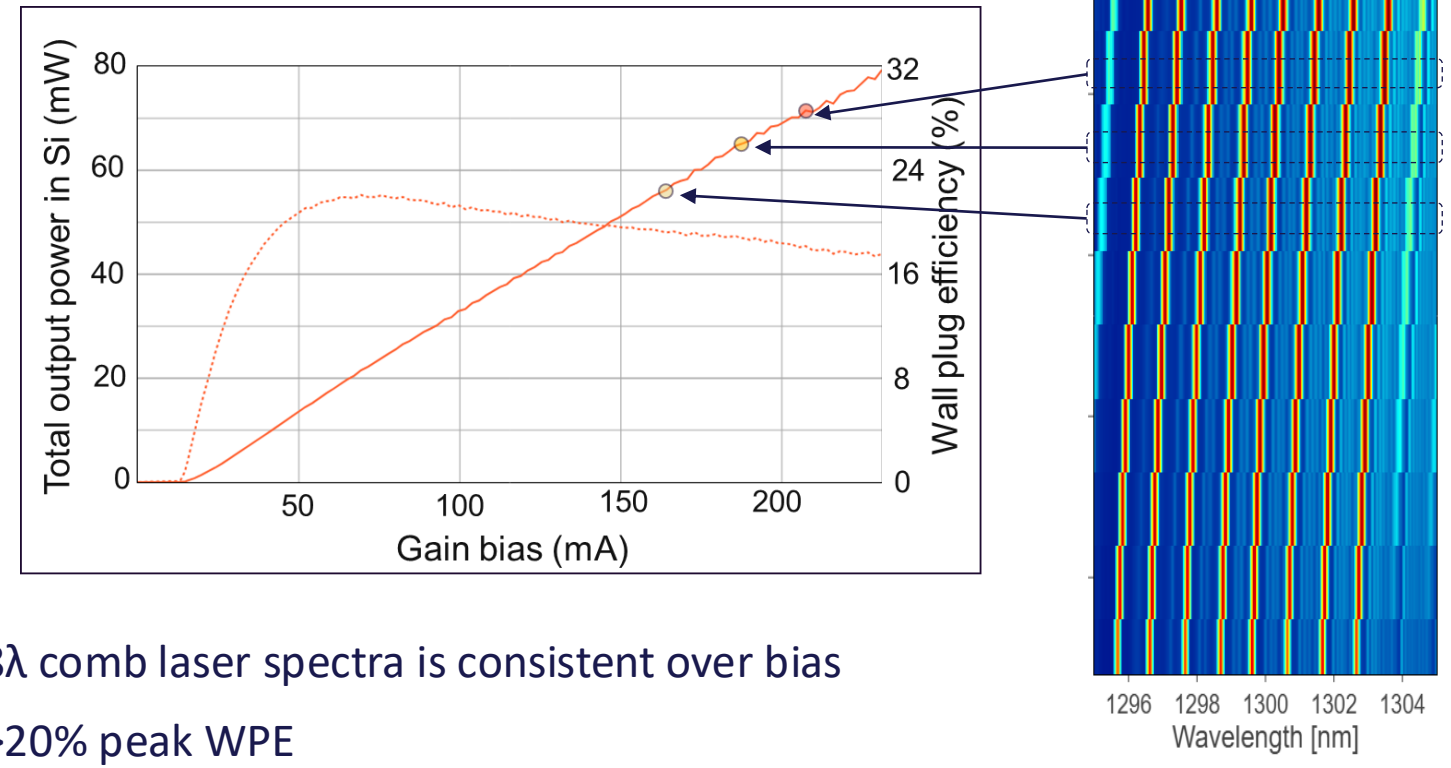
100°C



110°C



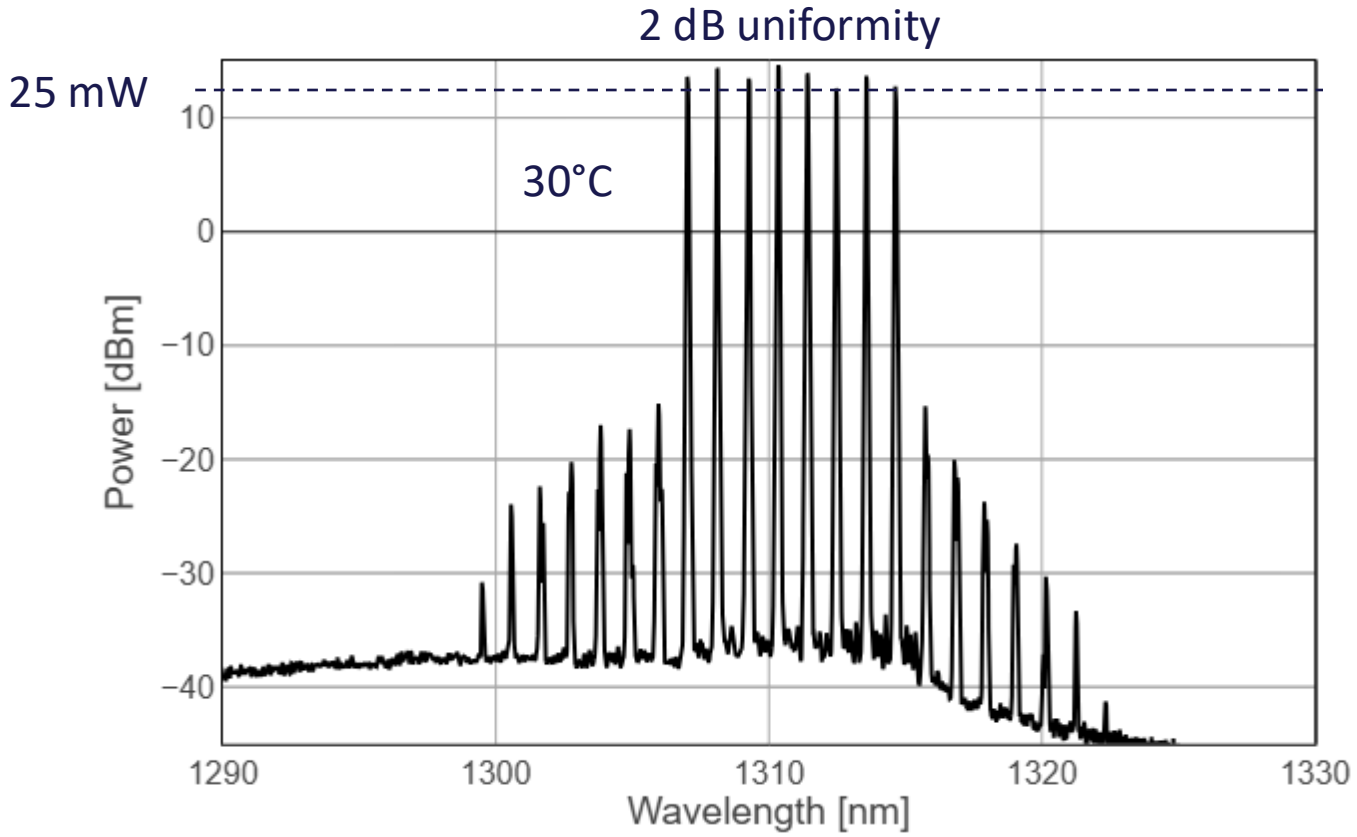
Stable Comb Spectra vs Bias



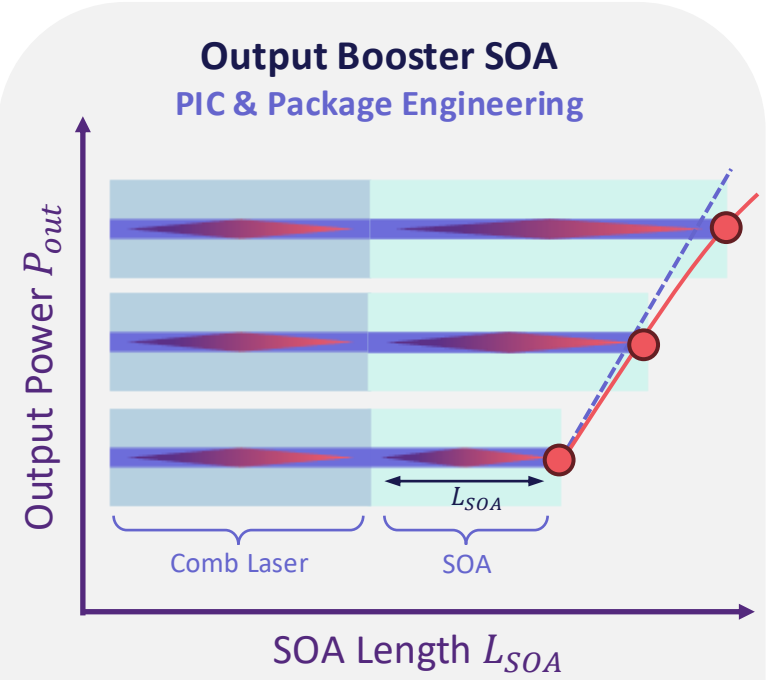
- 8λ comb laser spectra is consistent over bias
- >20% peak WPE
- Achieving >80 mW total output power in the Si waveguide from comb laser alone

NOTE: Jagged edges between bias points are not mode hops, they are an artifact of the contour plot.

200 mW 8λ comb + SOA source (>25 mW / λ)



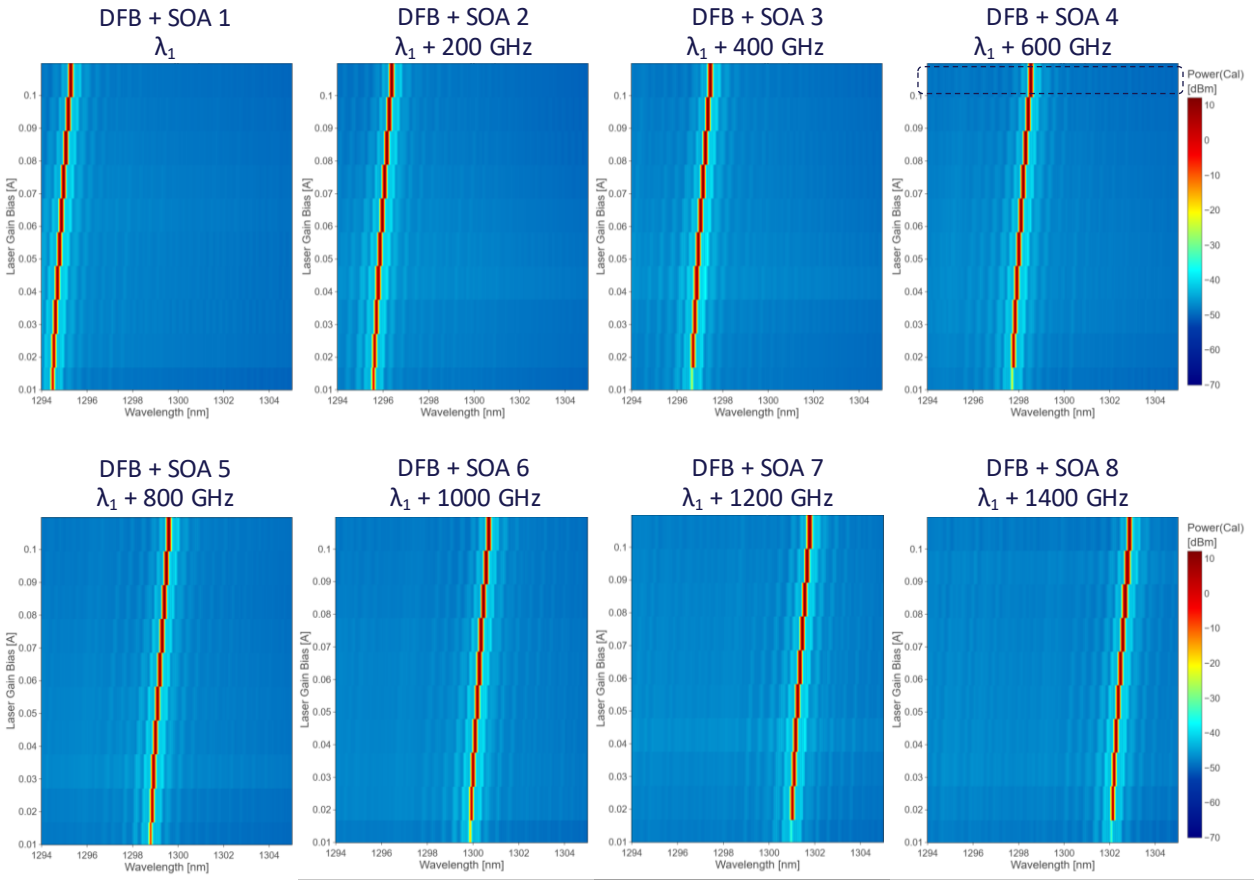
Out of band SMSR will be improved with coupling loss reduction and SOA optimization.



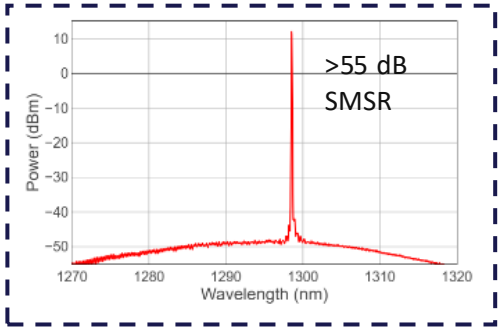
Increased output power with booster SOA L_{SOA}; integrated on Si or external III-V

High Performance Single-λ DFBs on Silicon

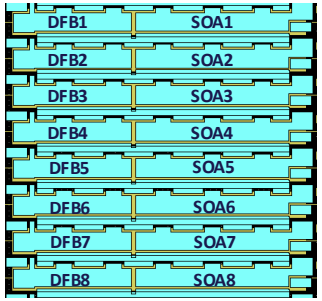
Optical spectra vs. laser bias y-axis, SOA bias fixed at 150 mA



NOTE: Jagged edges between bias points are not mode hops, they are an artifact of the contour plot.



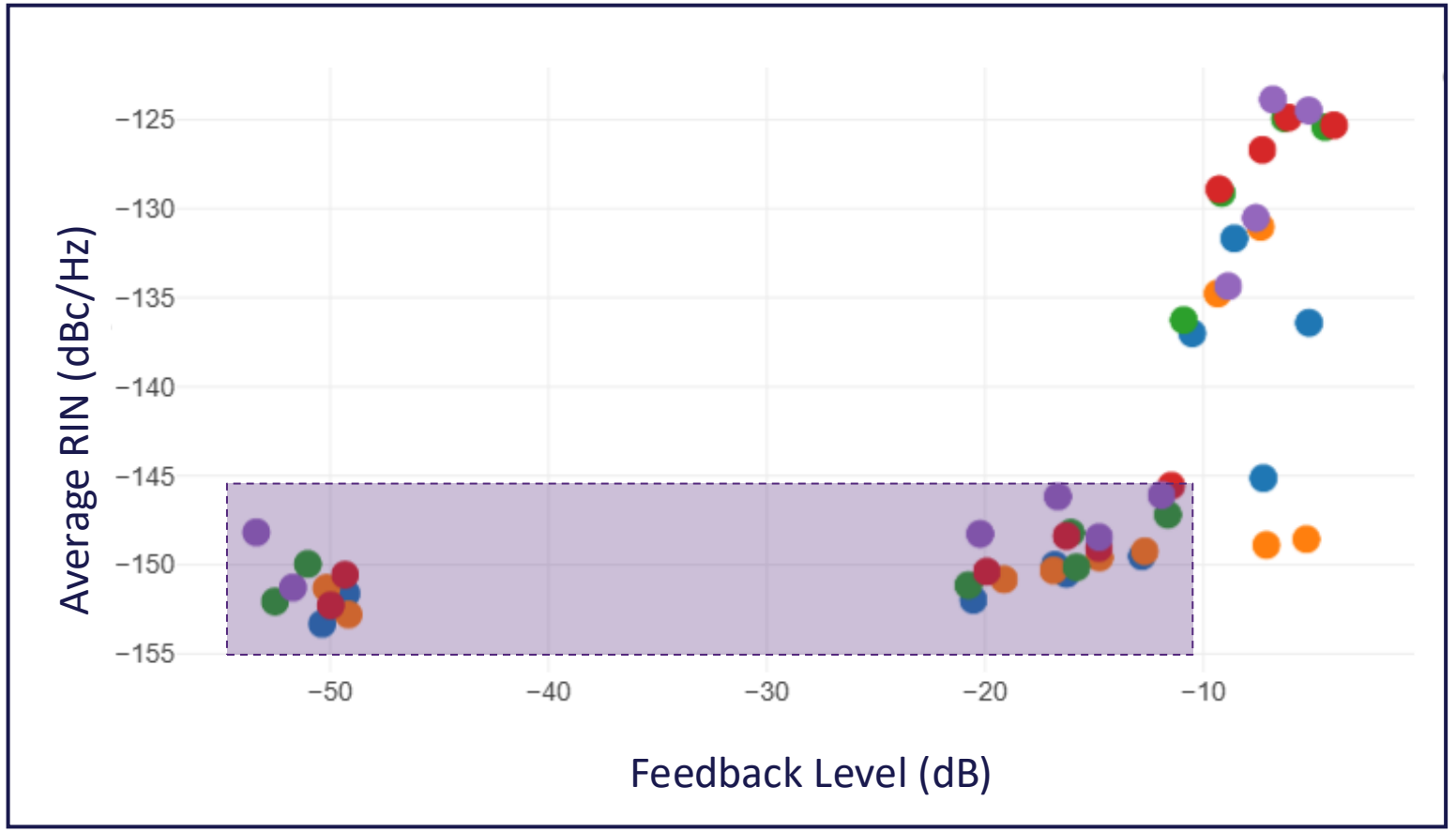
8x DFB + SOA Array Layout



- Demonstrated 8x200 GHz DFB+SOA arrays on Si
  - High SMSR, mode hop free
  - Up to 85 mW single side power in Si WG
  - Precise spacing
  - High tolerance to back reflection (without SOA)
  - Not yet optimized for power or efficiency.

QD DFBs stable to >15 dB of feedback

- 5 different wavelength DFB+SOAs with SOA biased at material transparency (no SOA gain)
- RIN stays near −150 dBc/Hz for optical feedback up to −15 dB
- QD DFB can likely operate isolator free when coupled to a PIC



Summary and Next Steps

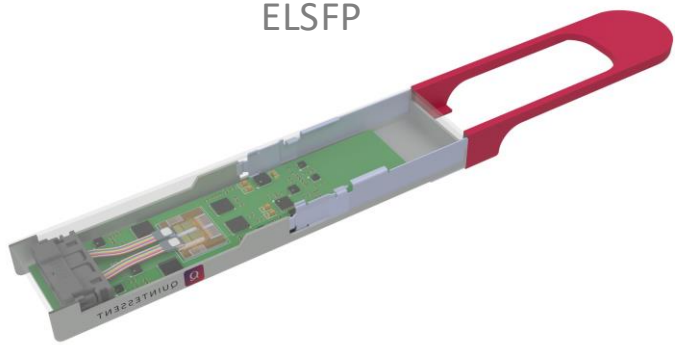
Summary

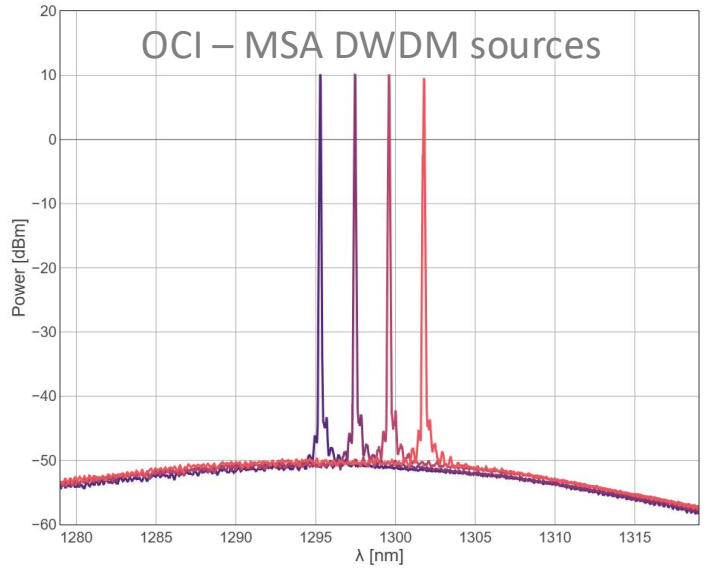
- O-Band Quantum Dot on Si platform eliminates multiple supply chain chokepoints (InP, e-beam, regrowth, isolators)
- Single and multi-wavelength lasers possible with the same platform

Next

- OCI-MSA sources: 4λ × 400 GHz per group, one comb per group
- ELSFP form factor
- More power per λ
- More reliability testing and validation

ELSFP





OCP FUTURE TECHNOLOGIES SYMPOSIUM

October 14, 2026
San Jose, California

